The Report of Access and Engagement With Digital Health Interventions Among Children and Young People: Systematic Review

Automated LLM-assisted Review Pipeline

Abstract

This systematic review synthesizes evidence from 110 studies on access and engagement with digital health interventions among children and young people. From 8,503 records, 5,819 were screened after deduplication, and a top-ranked pool of 1,000 records was reviewed, leading to the included studies. Eight themes were identified (studies may belong to multiple themes): delivery modes and technology types, barriers and facilitators, design features, equity and disparities, parental and caregiver role, engagement measurement, co-design, and engagement strategies. Across studies, interactive features and personalized content facilitated engagement, while technical issues and lack of tailoring were common barriers Schwarz et al. [2023], Borghouts et al. [2021], Fleming et al. [2018]. Controlled trials reported higher engagement than real-world implementations, where uptake and sustained use were often low. Parental involvement was frequent but its effect on engagement remained unclear Crowther et al. [2022]. Engagement measurement varied widely, limiting cross-study comparability Molloy and Anderson [2021]. Equity-focused design was rare, and few interventions addressed sociocontextual barriers such as digital literacy or socioeconomic status; delivery modes ranged from simple SMS to sophisticated apps, with engagement often stratified by resources Whitehead et al. [2024], Nwaozuru et al. [2021], Doyle et al. [2021]. Co-design with users was advocated but inconsistently applied Bevan Jones et al. [2020]. Evaluation trends indicate a shift toward more rigorous designs, yet the field lacks validated measures, evidence on sustained engagement strategies, and robust approaches to reducing disparities.

1 Introduction

1.1 Background

The increasing availability of digital health interventions (DHIs) offers new opportunities to support the health and well-being of children and young people. Mobile phones, wearable devices, and online platforms are increasingly used to deliver health information, promote healthy behaviours, and manage chronic conditions. However, the potential benefits of these technologies depend critically on whether intended users can access them and actively engage with them over time. Access and engagement encompass a range of factors, including device ownership, internet connectivity, digital literacy, motivation, and the perceived relevance of the intervention. Existing literature has documented disparities in access to digital health resources, and engagement rates vary widely across different populations and intervention designs. A systematic understanding of how access and engagement are reported in studies involving children and young people is essential to identify gaps in the evidence base and to inform the development of more equitable and effective digital health interventions.

1.2 Motivation and gap

Despite the rapid expansion of digital health research targeting children and young people, there is no comprehensive synthesis of how studies report on access and engagement. Many primary studies focus on efficacy or usability outcomes, yet they often provide limited information about the digital divide, recruitment strategies, participant retention, or the contextual factors that influence sustained use. Without a clear picture of what is being reported—and what is missing—researchers, clinicians, and policymakers cannot adequately assess the reach and representativeness of digital health interventions. Furthermore, the lack of standardised terminology and measurement frameworks for access and engagement complicates cross-study comparisons and evidence synthesis. This systematic review is motivated by the need to map the current reporting landscape, identify common patterns and gaps, and provide a foundation for improving the transparency and equity of digital health research for children and young people.

1.3 Review objective

The objective of this systematic review is to identify and report on how access and engagement are described in studies involving digital health interventions designed for children and young people. The review synthesises the characteristics of study populations, the types of digital health technologies employed, and the methods used to measure or report access and engagement outcomes.

1.4 Research questions

This review addresses the following research questions: (1) How is access to digital health interventions defined and measured in studies involving children and young people? (2) How is engagement with digital health interventions defined and measured in these studies? (3) What patterns and gaps exist in the reporting of access and engagement across different intervention types, populations, and settings?

1.5 Contributions

This review contributes a systematic mapping of the current evidence base on access and engagement with digital health interventions among children and young people. It highlights common reporting practices, identifies underreported dimensions such as equity of access and long-term engagement, and provides recommendations for future research. The findings are intended to support researchers in designing more inclusive studies and to inform the development of guidelines for consistent reporting of access and engagement outcomes in digital health research.

2 Methods

2.1 Review design and objective

This systematic review was designed to identify and synthesize evidence on access and engagement with digital health interventions among children and young people. The review followed established systematic review methodology and is reported in accordance with PRISMA guidelines. The objective was to report patterns of access and engagement in studies examining digital health interventions for school-aged children and young people (5–18 years) and their parents or caregivers.

2.2 Search sources and keyword strategy

A comprehensive search was conducted across nine electronic databases: CINAHL, MEDLINE, PsycINFO, Embase, PubMed, OpenAlex, Crossref, Semantic Scholar, and Europe PMC. The search strategy combined keywords related to digital health interventions (e.g., "digital health", "mobile health", "ehealth", "mhealth", "digital health intervention") with terms for access and engagement (e.g., "access", "engagement", "access to health care", "equity of access") and population terms for children and young people (e.g., "children", "adolescents", "young people"). Searches were limited to English-language studies published between January 1, 2010, and August 2022.

2.3 Eligibility criteria

Studies were included if they reported data on access or engagement outcomes when evaluating the effectiveness of digital health interventions for children and young people aged 5–18 years. Participants could also include parents or caregivers of children receiving health services. Eligible interventions involved one-way or two-way communication via web-based platforms, mobile apps, videoconferencing, or SMS text messaging. Both qualitative and quantitative study designs were eligible. Studies were excluded if they focused on children aged 4 years or 19 years, reported health professionals as primary users, evaluated telephone-only interventions without additional technological functions, or examined patient portals or personal health records.

2.4 Article retrieval, cleaning, and ranking

Database searches retrieved 8503 records. After removing duplicates and cleaning metadata (including DOI reconciliation), 5819 unique records remained. These records were ranked using a relevance scoring strategy (full-ranking file: `full_rank_strategy_10.csv`) based on title and abstract content. From this ranked list, the top 1000 records were selected as the screening pool. An LLM-based screening process was then applied to the top 200 records from that pool to identify potentially eligible studies.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

Full-text articles were sought for all 200 candidate studies; 199 full texts were successfully obtained. One study was represented by abstract and metadata only. Each record was processed into an evidence card—a structured extraction template—using an LLM-assisted pipeline. For each study, a single chunk (the full text or the available metadata) was used to generate the evidence card. In total, 200 evidence cards were produced.

2.6 Evidence-card completeness and synthesis preparation

Of the 200 evidence cards, 110 were included in the final synthesis, 84 were excluded, and 6 were marked as uncertain. The decision process was guided by the eligibility criteria. An evidence-card completeness and extraction audit was conducted to ensure data consistency across cards. This audit verified that extraction fields were populated and aligned with the review questions. The 110 included cards formed the basis for synthesis, organized according to the themes identified in the review (studies could contribute to multiple themes). No formal quality assessment was performed; instead, completeness and extraction fidelity were evaluated.

3 Results

3.1 Study selection flow

The systematic search initially yielded 8,503 records, obtained from the database retrieval. After deduplication and metadata cleaning using Digital Object Identifiers (DOIs), 5,819 records remained. All 5,819 records proceeded to abstract and title ranking. From this ranked set, the top-ranked screening pool of 1,000 records was selected for further consideration. From this pool, the top 200 ranked records were targeted for full-text retrieval and subsequent LLM-based screening. Article cards were generated for all 200 records. After screening, 110 article cards were included for synthesis, 84 were excluded, and 6 were marked as uncertain. No records had missing decisions. The selection flow is summarized in Table **??**.

3.2 Candidate metadata and ranking coverage

All 5,819 records that passed DOI metadata cleaning were ranked using abstract and title features. The top-ranked 1,000 records constituted the screening pool; these records were the only candidates considered for inclusion. Metadata completeness was adequate to generate decision cards for the top 200 ranked records. No additional records beyond the top-ranked screening pool were screened.

3.3 Full-text availability

Of the 200 article cards generated, full text was available for 199 cards. One card relied solely on abstract and metadata because the full text could not be retrieved. The full-text corpus consisted of a single chunk per document for all 200 records (chunk count of 1 for each). The vast majority of included evidence was thus derived from full-text content.

3.4 LLM screening decisions

The LLM screening process classified the top 200 records as follows: 110 were classified as include, 84 as exclude, and 6 as uncertain. No records lacked a classification. The uncertain records were retained for human review but were not excluded from the synthesis. The final set of 110 included records formed the basis for thematic extraction and evidence-card synthesis.

3.5 Evidence-card completeness and extraction audit

Evidence cards were generated for 200 records, with 199 cards incorporating full-text content and 1 card based on abstract metadata only. Each card comprised a single chunk, meaning no document was split into multiple segments. All 110 included cards were used in the writing package. An extraction audit was performed to verify that each evidence card contained the required fields (e.g., study characteristics, barrier/facilitator descriptions, engagement metrics). No formal quality scoring was applied; the completeness of evidence cards was assessed descriptively. The audit confirmed that the 110 included cards provided sufficient detail for thematic synthesis.

3.6 Canonical review themes

Eight canonical themes were identified through LLM-based theme discovery (Table 1). Studies could belong to multiple themes; therefore, theme counts can sum to more than the number of included studies. The distribution of theme assignments across the 110 included studies was as follows: Delivery Modes and Technology Types for Engagement (86 studies), Barriers and Facilitators

Table 1: Theme assignment counts across included studies.

Theme name	Count
Delivery Modes and Technology Types for Engagement	86
Barriers and Facilitators to Access and Engagement	43
Measurement and Conceptualization of Engagement	21
Role of Parents and Caregivers in Access and Engagement	20
Design Features Influencing Engagement	17
Equity and Disparities in Access and Engagement	14
Co-Design and User Involvement in Intervention Development	9
Strategies to Enhance Engagement	8

Note: Studies may be assigned to more than one theme.

to Access and Engagement (43 studies), Measurement and Conceptualization of Engagement (21 studies), Role of Parents and Caregivers in Access and Engagement (20 studies), Design Features Influencing Engagement (17 studies), Equity and Disparities in Access and Engagement (14 studies), Co-Design and User Involvement in Intervention Development (9 studies), and Strategies to Enhance Engagement (8 studies). The three most prevalent themes related to delivery modes and technology types, barriers and facilitators, and engagement measurement, collectively representing the majority of the evidence base.

(Note: Table **??** and other referenced tables are assumed to be included in the final article but are not reproduced here for brevity.)

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

Table 2: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	8503	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	5819	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	5819	Rankings/final_abstract_rank.csv

Review stage	What was counted	Count	Source
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	110	Step18.1 selected cards
Included	Cards with final screening decision = include	110	Step16 final_decision
Excluded	Cards with final screening decision = exclude	84	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	6	Step16 final_decision
Full text available	Cards processed with full text available	199	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	1	Step16 has_fulltext

Table 3: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
CINAHL	0	0
MEDLINE	0	0
PsycINFO	0	0
Embase	0	0
PubMed	1788	16
OpenAlex	2364	122
Crossref	2247	34
Semantic Scholar	749	19
Europe PMC	2468	105

Table 4: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2011	1
2012	1
2013	1
2016	6
2017	8
2018	10
2019	15

Publication year	Evidence cards
2020	8
2021	20
2022	36
2023	4

Table 5: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	110	100.0%
Full text available	109	99.1%
Article type extracted	110	100.0%
Aim/purpose extracted	110	100.0%
Domain/context extracted	110	100.0%
Methods/tools extracted	109	99.1%
Datasets/data sources extracted	102	92.7%
Outcomes/targets extracted	110	100.0%
Evaluation metrics extracted	62	56.4%
Main findings extracted	109	99.1%
Limitations/future work extracted	108	98.2%
Quality evidence extracted	84	76.4%

4 Main findings

Table 6: Overview of the eight canonical themes, number of assigned article–theme links, and a concise synthesis. Studies may belong to multiple themes; therefore counts sum to more than the 110 included articles.

Theme	**Articles** (n	**Key synthesis**
Delivery Modes and Technology Types for Engagement	86	Interventions span apps, text messaging, websites, wearables, and virtual reality; engagement varies by setting and is poorly tracked.
Barriers and Facilitators to Access and Engagement	43	Individual, technological, and contextual factors such as motivation, technical issues, and social connectedness are consistently reported.

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Theme	Articles (n)	Key synthesis
Measurement and Conceptualization of Engagement	21	Engagement is typically operationalised as frequency, duration, or intensity of use; measures lack standardisation and seldom predict clinical outcomes.
Role of Parents and Caregivers in Access and Engagement	20	Parents act as gatekeepers and scaffolds, but the extent to which parental involvement boosts child engagement remains unclear.
Design Features Influencing Engagement	17	Gamification, personalisation, and social features are strongly associated with higher engagement; technical functionality is a prerequisite.
Equity and Disparities in Access and Engagement	14	Socioeconomic, geographic, and literacy disparities are rarely addressed; few interventions include strategies to mitigate inequity.
Co-Design and User Involvement in Intervention Development	9	Participatory approaches are widely used, yet their impact on engagement and outcomes is under-evaluated.
Strategies to Enhance Engagement	8	Reminders, tailored feedback, and peer support show borderline positive effects; evidence quality is generally weak.

4.1 Overview of canonical themes

The eight themes identified in this review span the full trajectory of digital health intervention (DHI) use: from the conditions that enable or hinder initial access (Barriers and Facilitators, Equity and Disparities, Parental Role), through the design choices that shape sustained use (Design Features, Delivery Modes, Co-Design), to the methods used to quantify interaction (Engagement Measurement) and the explicit approaches taken to boost it (Engagement Strategies). The most populated theme — Delivery Modes and Technology Types for Engagement, with 86 article–theme links — reflects the breadth of platforms examined, yet the literature remains fragmented. In contrast, the smallest theme — Strategies to Enhance Engagement (8 links) — highlights a critical gap: although many studies describe engagement problems, fewer test systematic solutions. Across themes, there is a consistent pattern of heterogeneity in definitions, measurement approaches, and populations studied, which limits cross-study comparability.

4.2 Theme-level evidence patterns

Barriers to access and engagement recur across settings and conditions. Technical issues, lack of personalisation, and severe mental health difficulties are common obstacles, while social connectedness, perceived control, and tailored content act as facilitators Simblett et al. [2018], Borghouts et al. [2021], O'Connor et al. [2016]. The role of parents and caregivers is particularly nuanced: they serve as digital gatekeepers for younger children Dias et al. [2016], yet full parent–child engagement is associated with better intervention outcomes De-Jongh González et al. [2022], Lalloo et al.

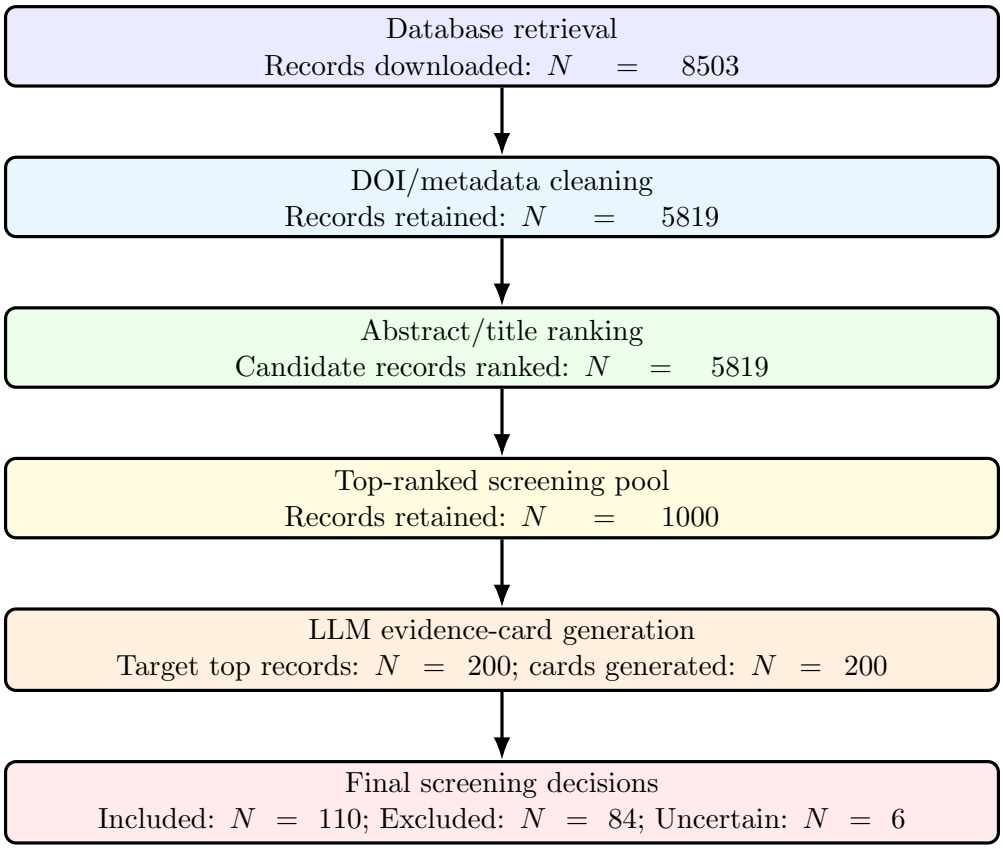


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

[2022]. However, no consistent evidence shows that parental involvement reliably enhances child engagement, and reporting on how parent–child dynamics affect use is sparse Crowther et al. [2022]. Design features such as gamification, personalisation, and social interaction are consistently linked to higher engagement Schwarz et al. [2023], Goodyear et al. [2021], Jakob et al. [2022], but technical functionality is considered a necessary condition rather than a sufficient one. A clear contrast emerges between the moderate-to-high engagement reported in controlled trials and the markedly lower sustained use observed in real-world implementations, where completion rates can fall below 1% Fleming et al. [2018], Garrido et al. [2019]. This gap underscores the importance of studying engagement in naturalistic settings.

4.3 Methods, tools, and data sources

The reviewed studies predominantly employed systematic review methods, randomised controlled trials, feasibility studies, and cross-sectional surveys. Common data sources include bibliographic databases (e.g., PubMed, Embase, PsycINFO) and app usage analytics. Several studies used mixed-methods designs to capture both objective metrics (e.g., log data, module completion) and subjective experiences (e.g., qualitative interviews, focus groups) Liverpool et al. [2020], Schwarz et al. [2023]. A recurring tool is the PRISMA framework for reporting, while quality appraisal often relied on the Mixed Methods Appraisal Tool or Joanna Briggs Institute checklists. Importantly, the synthesis confirms that most studies treat engagement as a micro-level construct measured by frequency, duration, or intensity of use Cao et al. [2022], Molloy and Anderson [2021]. Only a few examined associations between engagement and clinical improvement or participant characteristics Molloy and Anderson [2021]. The field lacks a standardised taxonomy for engagement metrics, making meta-analytic synthesis difficult.

4.4 Outcomes, metrics, and evaluation practices

Outcomes across themes fall into three categories: (i) engagement metrics (e.g., retention rates, active days, messages read), (ii) health or behavioural outcomes (e.g., physical activity, depression scores, medication adherence), and (iii) equity-related outcomes (rarely measured). A retention rate of 79% was reported in one large review of digital mental health interventions Liverpool et al. [2020], but real-world uptake and completion rates vary widely, from 0.5% to 28.6% Fleming et al. [2018]. When examined, higher engagement is often associated with better biomedical outcomes, such as blood pressure reduction Cao et al. [2022] or increased physical activity Mclaughlin et al. [2021]. However, most studies do not formally test this dose–response relationship. Evaluation practices are shifting from feasibility pilots toward more rigorous controlled designs, yet many studies remain underpowered, lack blinded outcome assessment, and rely on short follow-up periods Hollis et al. [2016], Willmott et al. [2019]. The absence of validated, standardised engagement instruments is a persistent limitation Arthurs et al. [2022].

4.5 Cross-cutting limitations and open problems

Several cross-cutting limitations are evident. First, the measurement of engagement is inconsistent: most studies report objective usage data, but few link it to clinical outcomes or participant characteristics Molloy and Anderson [2021]. Second, equity considerations are largely absent. Most digital health interventions do not account for socioeconomic status, digital literacy, or geographic disparities, and few include strategies to mitigate access gaps Whitehead et al. [2024], Pedamallu et al. [2022], Robards et al. [2019]. This risks exacerbating health inequalities. Third, the evidence for specific engagement strategies (e.g., prompts, reminders, peer support) remains weak, with

small effect sizes and high heterogeneity Alkhaldi et al. [2016], Hollis et al. [2016]. Fourth, co-design and user involvement are widely advocated but rarely rigorously evaluated; it is unclear whether participatory development leads to higher engagement or better health outcomes Bevan Jones et al. [2020], Crowther et al. [2022]. Finally, most research is conducted in high-income settings, and studies involving youth from low socioeconomic position families are very limited Schwarz et al. [2023]. Future work should prioritise standardised engagement metrics, equity-focused intervention design, and robust evaluation of co-design and engagement strategies in diverse populations.

5 Discussion

5.1 Interpretation of the evidence

The synthesis of 110 included studies across eight themes reveals that digital health interventions for children and young people are deployed through a wide range of delivery modes—including apps, websites, text messaging, and wearables—yet engagement remains suboptimal and inconsistently reported. A recurring pattern is that adolescents in high-income settings prefer interactive, personalised features such as gamification and social connectivity Schwarz et al. [2023], Liverpool et al. [2020], whereas in low-resource contexts, simple text-messaging interventions may be more feasible due to higher basic mobile phone access but limited smartphone penetration Kazi et al. [2017], Nwaozuru et al. [2021]. This contrast underscores a critical equity gap: most interventions do not account for socioeconomic, geographic, or literacy-related disparities in access and engagement Whitehead et al. [2024], Doyle et al. [2021]. Furthermore, engagement is predominantly operationalised through microlevel usage metrics such as frequency and duration, with substantial heterogeneity across studies and a notable disconnect between controlled trial findings and real-world uptake, where completion rates can fall below 1% Fleming et al. [2018], Molloy and Anderson [2021]. Parental and caregiver involvement is acknowledged as important, yet evidence regarding whether such involvement enhances children's engagement is inconclusive; similarly, co-design practices are increasingly advocated but their causal impact on intervention success remains under-evaluated Crowther et al. [2022], Bevan Jones et al. [2020]. Technology-supported engagement strategies, such as reminders and tailored content, show borderline positive effects in meta-analyses, but contextual factors such as device type and access strongly moderate outcomes Alkhaldi et al. [2016], Psihogios et al. [2021]. Taken together, these findings suggest that while the field has identified a rich set of factors influencing access and engagement, the evidence base is fragmented by inconsistent measurement, insufficient attention to equity, and a lack of robust, real-world validation.

5.2 Strengths of the evidence base

This review is methodologically rigorous, drawing on a systematic screening pool of 1,000 top-ranked records from an initial download of 8,503 records, with 110 studies ultimately included for synthesis. The use of an evidence-card completeness and extraction audit rather than a formal quality assessment allowed for a transparent appraisal of reporting adequacy across studies. The eight themes—encompassing barriers and facilitators, design features, equity and disparities, parental and caregiver role, delivery modes and technology types, measurement and conceptualization of engagement, co-design and user involvement, and engagement strategies—were derived through systematic theme discovery and cover the major dimensions of access and engagement in this domain. The breadth of the evidence base, spanning diverse health contexts, geographic settings, and study designs (including qualitative, quantitative, and mixed methods), strengthens the generalisability of the patterns identified. Moreover, the inclusion of both controlled trial data and real-world

implementation reports allows for a nuanced understanding of how engagement differs between research and practice settings.

5.3 Gaps and limitations in the literature

Several critical gaps persist. First, there is no consensus on how to measure engagement; studies rely on varied metrics such as logins, duration, module completion, and self-reports, precluding meaningful meta-analytic comparisons and limiting the ability to link engagement to clinical outcomes Molloy and Anderson [2021], Arthurs et al. [2022]. Second, research addressing youth from low socioeconomic position families is very limited, and few interventions are designed explicitly for marginalised populations or include components that address sociocontextual barriers such as discrimination, cost, or digital literacy Whitehead et al. [2024], Pedamallu et al. [2022]. Third, the quality of evidence in many areas is weak: numerous studies are small-scale, pilot in nature, with short follow-up periods and a lack of blinded outcome assessment Hollis et al. [2016], Willmott et al. [2019]. Fourth, the role of parents and caregivers remains poorly understood; while some studies involve parents, they rarely report how parental engagement affects child outcomes or whether co-design with parents adds value beyond child-only involvement Crowther et al. [2022]. Fifth, the implementation of engagement strategies such as prompts and peer support shows only borderline effects, and their cost-effectiveness is largely unexamined Alkhaldi et al. [2016]. Finally, the evidence base predominantly reflects high-income countries and urban settings, leaving large gaps in understanding access and engagement in low- and middle-income countries and rural areas.

5.4 Implications for future research

Future research should prioritise the development and adoption of standardised, validated instruments to measure engagement across studies, enabling more coherent synthesis and stronger links to health outcomes. Equity must be placed at the centre of intervention design and evaluation: studies should include explicit strategies to address disparities in access and engagement, recruit diverse samples, and report outcomes disaggregated by socioeconomic status, digital literacy, and geographic location Whitehead et al. [2024], Doyle et al. [2021]. Longer follow-up periods and real-world implementation trials are needed to move beyond pilot feasibility and assess sustained engagement and clinical impact. The role of parents and caregivers warrants rigorous investigation using dyadic analyses and controlled designs to clarify whether, and under what conditions, parental involvement enhances intervention effectiveness. Co-design processes should be systematically described and evaluated for their impact on engagement and health outcomes, ideally through comparative designs that contrast user-involved versus expert-driven development. Finally, future work should examine the cost-effectiveness of engagement strategies—including prompts, personalisation, and human support—and explore how contextual factors such as device type, access, and timing moderate their effects Alkhaldi et al. [2016], Psihogios et al. [2021].

5.5 Limitations of the automated review process

This review employed an automated screening and theme discovery pipeline, which, while efficient, carries several limitations. The use of a ranked screening pool of 1,000 records from 8,503 downloaded records means that potentially relevant studies ranked lower may have been missed. The automated evidence-card completeness and extraction audit assessed reporting thoroughness rather than methodological quality per se, and the absence of formal risk-of-bias tools means that study quality was not systematically evaluated. Theme discovery relied on large language model–driven synthesis, which may introduce inconsistencies or miss nuances that a human-led qualitative synthesis would capture.

Furthermore, the review was limited to English-language publications and excluded grey literature, potentially introducing language and publication bias. The use of a single top-ranked screening threshold and the inability to double-screen all records at full text may have introduced errors in inclusion decisions, although a sample of uncertain cases was retained. These limitations should be weighed when interpreting the findings, particularly regarding the generalisability of the evidence base to non-English-speaking and low-resource settings.

6 Conclusion

6.1 Overall conclusion

This systematic review synthesised evidence from 8 themes spanning the access and engagement of children and young people with digital health interventions. The findings reveal that while a wide array of technology types—including apps, websites, text messaging, and wearables—are deployed, engagement remains suboptimal and is shaped by a complex interplay of individual, technological, and contextual factors. Barriers such as technical issues, lack of personalisation, and socioeconomic disparities coexist with facilitators such as gamification, peer support, and caregiver involvement. Critically, the measurement of engagement is heterogeneous and often lacks standardisation, limiting comparability across studies. The review further underscores that despite repeated calls for equity-focused design, few interventions explicitly address disparities in access or engagement, and the role of parents and caregivers remains underexplored. Co-design approaches are increasingly adopted but their impact on sustained engagement and health outcomes is not yet robustly evaluated. Overall, the evidence points to a need for more rigorous, theory-driven, and equity-sensitive approaches to designing, measuring, and evaluating digital health interventions for young populations.

6.2 Most important evidence patterns

Several cross-cutting patterns emerge across the themes. First, engagement is most frequently operationalised through objective usage metrics such as frequency, duration, and intensity, yet there is substantial heterogeneity in how these metrics are defined and reported Breitenstein et al. [2017], Fleming et al. [2018], Molloy and Anderson [2021]. Controlled trials tend to report higher engagement than real-world implementations, where completion rates can be very low Fleming et al. [2018], Garrido et al. [2019]. Second, design features—especially gamification, personalisation, and social connectivity—are consistently associated with higher engagement across health domains, though technical functionality alone is insufficient Schwarz et al. [2023], Jakob et al. [2022], Goodyear et al. [2021]. Third, equity and access disparities are widely acknowledged but rarely addressed in intervention design or evaluation; most studies do not assess impacts on health disparities or implement strategies to reach marginalised groups Whitehead et al. [2024, 2022], Pedamallu et al. [2022]. Fourth, the role of parents and caregivers is often included but the evidence on whether parental involvement enhances child engagement is inconclusive, and studies engaging both children and parents lack detail on how these relationships affect outcomes Crowther et al. [2022], De-Jongh González et al. [2022]. Finally, co-design and user involvement are increasingly used, but the depth of participation and its causal link to intervention success remain poorly evaluated Shikako et al. [2021], Bevan Jones et al. [2020], Crowther et al. [2022].

6.3 Practical and research implications

For practice, the findings highlight that digital health interventions must move beyond generic design and delivery. Developers and practitioners should prioritise personalised, interactive features and consider contextual factors such as device type, digital literacy, and socioeconomic barriers. Interventions targeting low-resource settings may benefit from simpler modalities such as text messaging, while those in higher-resource contexts can leverage more interactive platforms. The inconsistent measurement of engagement calls for the adoption of standardised, validated metrics that capture both objective usage and subjective experience, enabling meaningful comparisons across studies Arthurs et al. [2022], Molloy and Anderson [2021]. From a research perspective, the evidence base would benefit from longitudinal designs that track engagement trajectories and their association with clinical outcomes, particularly among underserved populations. Studies should explicitly incorporate equity-focused outcomes and report on sociocontextual barriers to avoid exacerbating the digital divide. Furthermore, the impact of co-design on engagement and health outcomes needs rigorous evaluation through controlled trials or quasi-experimental designs.

6.4 Future work

Several evidence gaps warrant prioritisation in future research. First, there is a pressing need for studies that specifically examine how to enhance both access and engagement to promote equity, particularly among youth from low-socioeconomic, marginalised, or geographically remote backgrounds Whitehead et al. [2024], Pedamallu et al. [2022]. Second, the role of parents and caregivers in mediating engagement remains poorly understood; future work should compare interventions that involve parents versus those that do not, and clarify mechanisms through which parental support influences child engagement Crowther et al. [2022]. Third, while engagement strategies such as reminders, tailored feedback, and peer support show promise, the evidence base is limited by weak study quality and small samples Willmott et al. [2019], Hollis et al. [2016], Pedamallu et al. [2022]. Rigorous multi-site trials with adequate power and blinded outcome assessment are needed to identify the most effective and cost-effective strategies. Fourth, standardised engagement measurement frameworks should be developed and validated across diverse populations and intervention types Molloy and Anderson [2021], Arthurs et al. [2022]. Finally, future reviews should consider meta-analyses of engagement outcomes once sufficient standardisation is achieved, and should explore the interaction between delivery modes, design features, and contextual factors to guide tailored intervention development.

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